Arav Palsule

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LinkedIn — GitHub — Portfolio

B.E. Data Science (2027) — CGPA: 8.34/10

Professional Summary

Aspiring Data Analyst with a strong foundation in statistics, data analysis, and quantitative problem-solving. Skilled in Python, SQL, data cleaning, exploratory data analysis (EDA), data visualization, and statistical analysis. Experienced in working with historical datasets, performing feature engineering, analyzing trends, building forecasting models, and evaluating results using statistical metrics. Interested in transforming raw data into actionable insights and supporting data-driven decision-making.

Education

A. P. Shah Institute of Technology, Thane 2023 – 2027
Bachelor of Engineering in Data Science **CGPA: 8.34/10**

Technical Skills

Programming: Python, SQL
Data Analysis: Data Cleaning, Data Wrangling, Exploratory Data Analysis (EDA), Data Preprocessing, Feature Engineering, Trend Analysis
Statistical Analysis: Descriptive Statistics, Statistical Analysis, Correlation Analysis, Regression Analysis, Hypothesis Testing
Python Libraries: Pandas, NumPy, Matplotlib, Seaborn
Databases: MySQL, SQLite, SQL Queries, Data Retrieval
Data Science: Predictive Analytics, Forecasting, Model Evaluation, MAE, RMSE, SMAPE
Tools and Platforms: Git, GitHub, Flask, Supabase, Excel

Projects

Dengue Outbreak Prediction – Data Analysis and Forecasting

- Prepared dengue case, weather, geographic, and temporal data for forecasting; performed feature engineering, lag analysis, and rolling-statistic calculations.
- Compared **four regression models**: Random Forest, Extra Trees, HistGradientBoosting, and Ridge Regression.
- Evaluated forecasts using chronological holdout and rolling-window validation with **MAE, RMSE, and SMAPE**.

Technologies Used: Python, Pandas, NumPy, Scikit-learn, Random Forest, Extra Trees, HistGradientBoosting, Ridge Regression

SPHERE AI – Educational Data Analytics Platform

- Developed an educational analytics platform for **three user roles**: students, teachers, and administrators.
- Built reporting workflows and dashboards to examine student progress, assessment performance, and classroom activity.
- Used academic KPIs and data visualizations to support analysis of education and engagement data.
- Connected assessment and document workflows with AI tutoring and learning-material retrieval.

Technologies Used: Python, Flask, Pandas, MySQL/SQLite, Plotly, OCR, FAISS, Sentence Transformers, RAG

The Review Room – Review and Text Analytics

- Built a **four-part NLP workflow** for review preprocessing, sentiment analysis, topic exploration, and review retrieval.
- Analyzed customer feedback to identify sentiment patterns, recurring topics, and aspect-level insights.
- Retrieved relevant reviews to support closer examination of the patterns found in the dataset.

Technologies Used: Python, NLP, Text Processing, Sentiment Analysis

CrackX – Road Condition Data Management

- Structured road-damage reports with location, image/video evidence, damage category, and severity information.
- Integrated **YOLOv8** detection outputs, including damage localization and severity assessment, into reporting workflows.
- Added duplicate-report checks and time-range reports for reviewing work across **five user roles**.

Technologies Used: Python, Pandas, SQL, YOLOv8, OpenCV, FastAPI, Supabase

Experience

AI Intern – Ennoble Social Innovation
September 2026 – October 2026

- Created custom Claude Skills using context engineering to automate business tasks, including PDF formatting according to internal document structures and standards.
- Developed an employee focus-monitoring prototype integrating face detection and emotion analysis to estimate focus scores and identify potential sleep or inactivity events.
- Designed automated screenshot capture and HR reporting workflows to communicate detected events and employee focus patterns.
- Built workflow automations using Zapier, LLMs, and API integrations to streamline repetitive operational processes.

Achievements

- **Rank 10** – C.R. Rao Statistics & Probability Olympiad: Ranked among 2.5 lakh+ participants across Asia.
- **AIR 3192, AICR 35** – Aryabhatta National Maths Competition 2026: Scored 100% in Round 1 and 80% in Round 2.
- **2nd Runner-Up** – QUASAR 4.0 Hackathon: Delivered a technical solution under a 24-hour development deadline.
- **Elite Rank** – NPTEL Multivariable Calculus: Recognized for mathematical and quantitative problem-solving skills.

Leadership

Founding President – Maths & Applied Science Club — 2026

- Founded and led the club, organizing 25+ technical events to promote mathematics, science, and technical learning among students.

Research Member – DSA Club — 2025 – 2026

- Explored data science, machine learning, and analytical workflows; mentored junior members in technical concepts.